HDOC Day-60

C Programming Activity | Anagram Checking Using Pointers

Problem Statement

Write a C program that reads two strings, s and t. Assume both strings contain only lowercase characters. Pass the strings to a function named check_anagram() using their addresses. Use pointers to pass the strings between the calling function and check_anagram().

The function check_anagram() must:

- Determine whether s and t contain the same characters with the same frequencies.
- Print "Anagram" if the two strings are valid anagrams.
- Print "Not Anagram" if the two strings are not valid anagrams.
- Treat the order of characters as irrelevant.
- Consider strings of different lengths as not anagrams.
- Use a character-frequency-counting approach to solve the problem.

C Program

```
#include <stdio.h>

void check_anagram(char *s, char *t)
{
   int freq[26] = {0};
   while (*s)
   {
      freq[*s - 'a']++;
      s++;
   }
   while (*t)
   {
      freq[*t - 'a']--;
      t++;
   }
   for (int i = 0; i < 26; i++)
   {
      if (freq[i] != 0)
      {
         printf("Not Anagram\n");
         return;
      }
   }
   printf("Anagram\n");
}
```

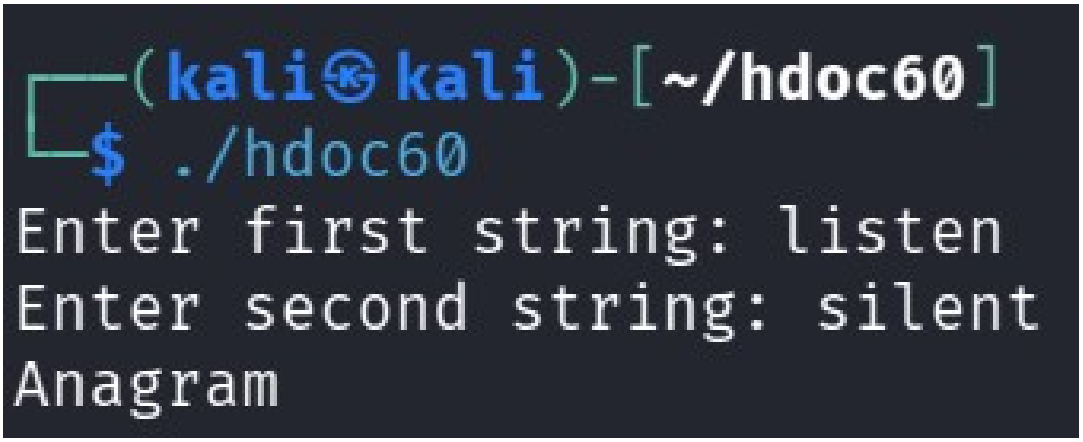
```
int main()
{
   char s[100], t[100];
   printf("Enter first string: ");
   scanf("%99s", s);
   printf("Enter second string: ");
   scanf("%99s", t);
   check_anagram(s, t);
   return 0;
}
```

Execution Screenshot

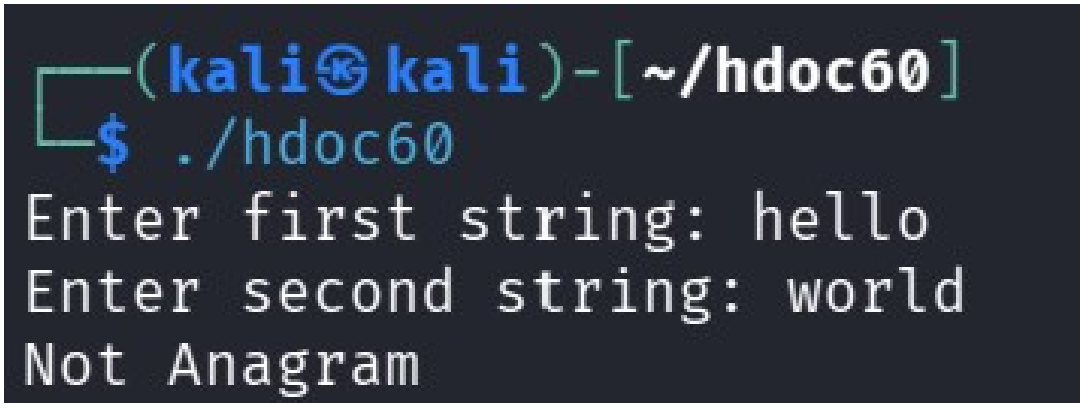
Compilation



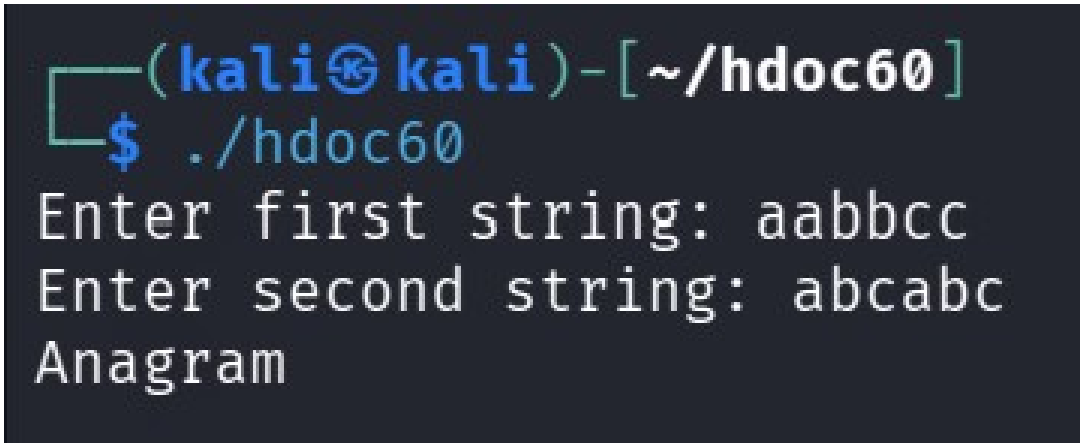
Execution 1 - Valid Anagram



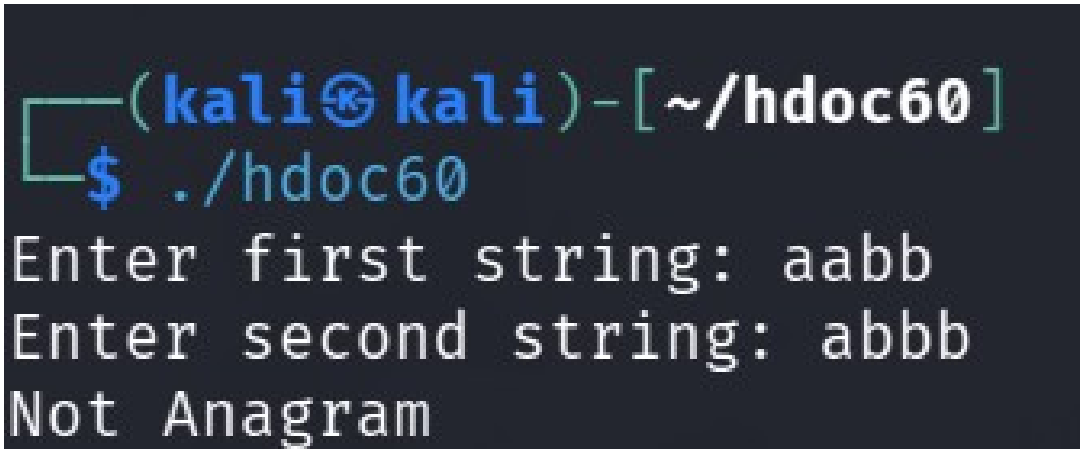
Execution 2 - Not Anagram



Execution 3 - Valid Anagram (Repeated Characters)



Execution 4 - Not Anagram (Different Frequencies)



Execution 5 - Not Anagram (Different Lengths)

